Simulation Lambda yield summary

trigger columns: % of beam-trigger reactions, each with KVC veto

yields x eff_tpc_L = 0.60; TPC: p and pi- also hit >= 5 TPC layers

reaction	beam	KVC veto	HTOF Mp	HTOF Fwd	phys.trig	TPC hit	yield	LH2 yield	TPC yield	TPC LH2
Lambda_eta	77959	98.91	48.14	63.14	66.58	48.76	3.336e+04	1.714e+04	2.444e+04	1.262e+04
Lambda_pi0	81087	99.14	31.84	43.97	49.08	40.13	6.163e+04	3.010e+04	5.042e+04	2.455e+04
Sigma0_pi0	81340	99.33	32.62	47.24	51.38	41.52	4.814e+04	2.355e+04	3.890e+04	1.907e+04
Sigmap_pim	80776	98.43	54.01	38.29	64.85	34.77	5.306e+04	2.557e+04	2.843e+04	1.387e+04
Lambda_pip_pim	80833	98.00	76.51	60.64	80.50	46.95	1.021e+05	5.019e+04	5.967e+04	2.932e+04
Sigma0_pip_pim	81045	97.99	75.74	64.40	80.56	47.56	2.086e+04	1.027e+04	1.233e+04	6.091e+03
Sigmap_pim_pi0	80733	98.82	52.93	49.27	69.22	35.94	1.723e+04	8.553e+03	8.973e+03	4.484e+03
Total							3.364e+05	1.654e+05	2.232e+05	1.100e+05

Lambda_eta : e72_tpcon-735_lambda_eta.root

Trigger condition (reaction events)	count	% of beam
saved reaction events	85078	109.13
kaon beam	77959	100.00
beam x KVC veto	77112	98.91
x HTOF Mp >= 2	37533	48.14
x HTOF forward	49225	63.14
x HTOF Mp >= 2 and forward	34853	44.71
physics trigger (Mp >= 2 or forward)	51905	66.58
x p and pi- TPC hits	38014	48.76
x <= 1 other TPC track (ntrack)	30152	38.68
x MM >= 0.5 GeV/c2	38014	48.76
x ntrack x MM >= 0.5 GeV/c2	30152	38.68
x no other TPC track (ntrack0)	26531	34.03
x kinematics	34068	43.70
x ntrack x MM x kinematics	27028	34.67

Generated reactions with beam trigger : 81242 (LH2 share 39278, 48.35 %)

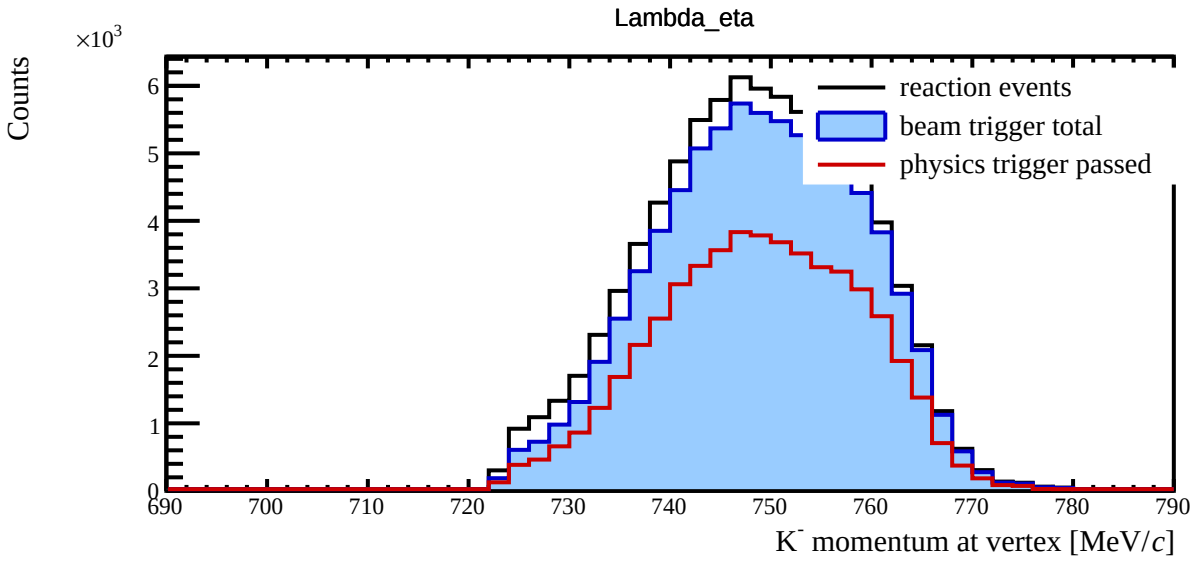
Vertex volume	beam trig.	phys. trig.	acc %	yield
LH2	39100	26295	67.25	1.714e+04
Kapton	3743	2522	67.38	1.627e+03
Mylar	1542	1039	67.38	6.722e+02
GFRP	27087	18093	66.80	1.168e+04
G10 top	4278	2616	61.15	1.492e+03
G10 bottom	2209	1340	60.66	7.512e+02

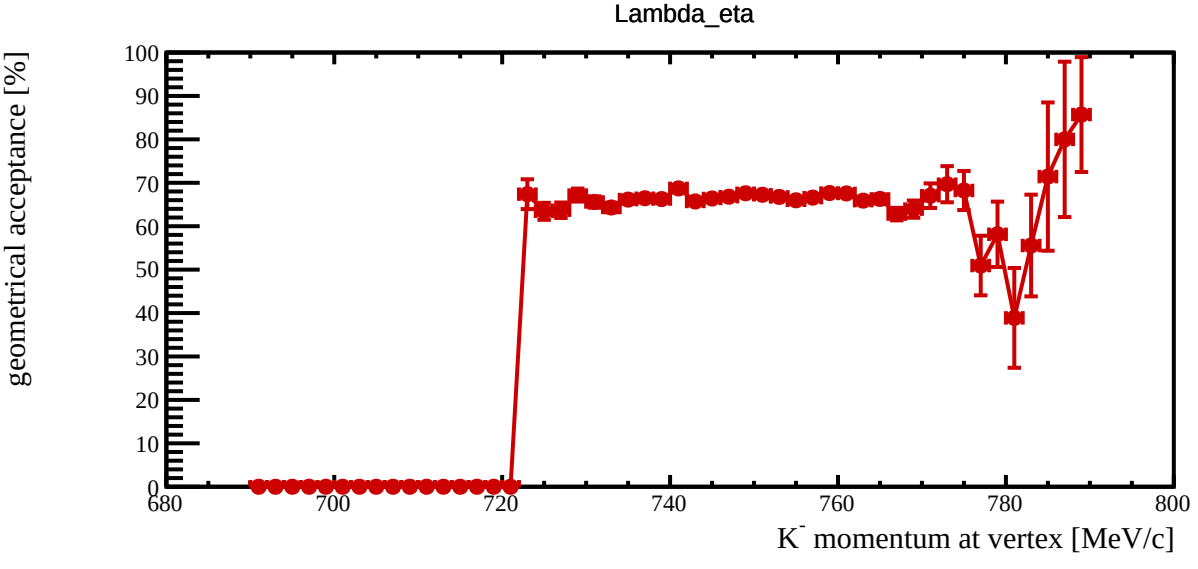
Expected Λ yield : 3.336e+04 (eff_tpc_L = 0.60)

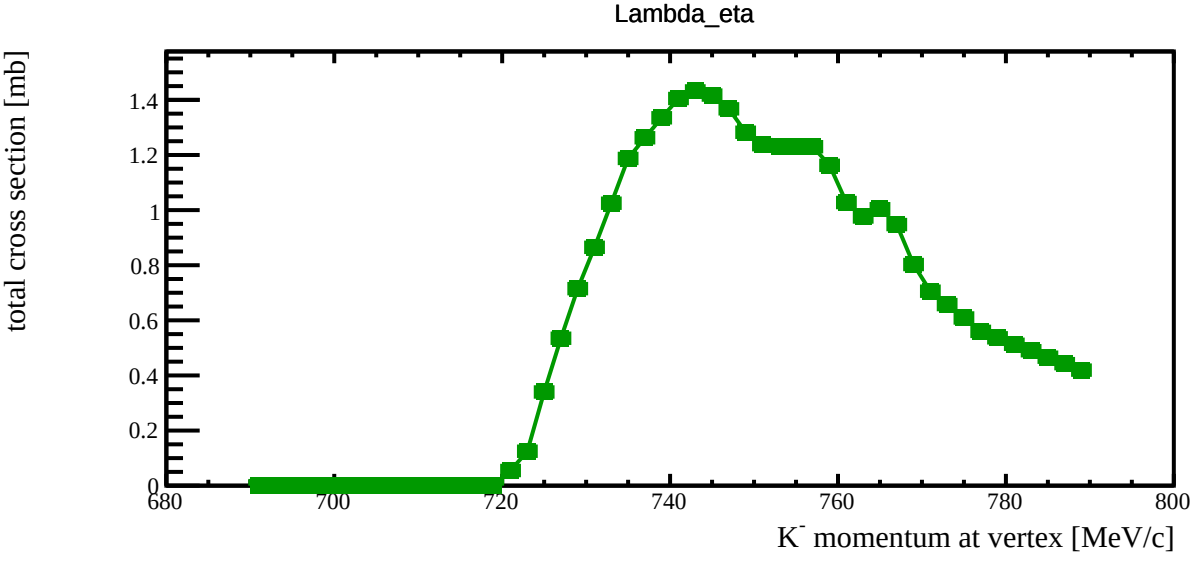
with p, pi- in >= 5 TPC layers : 2.444e+04 (also x eff_tpc_L)

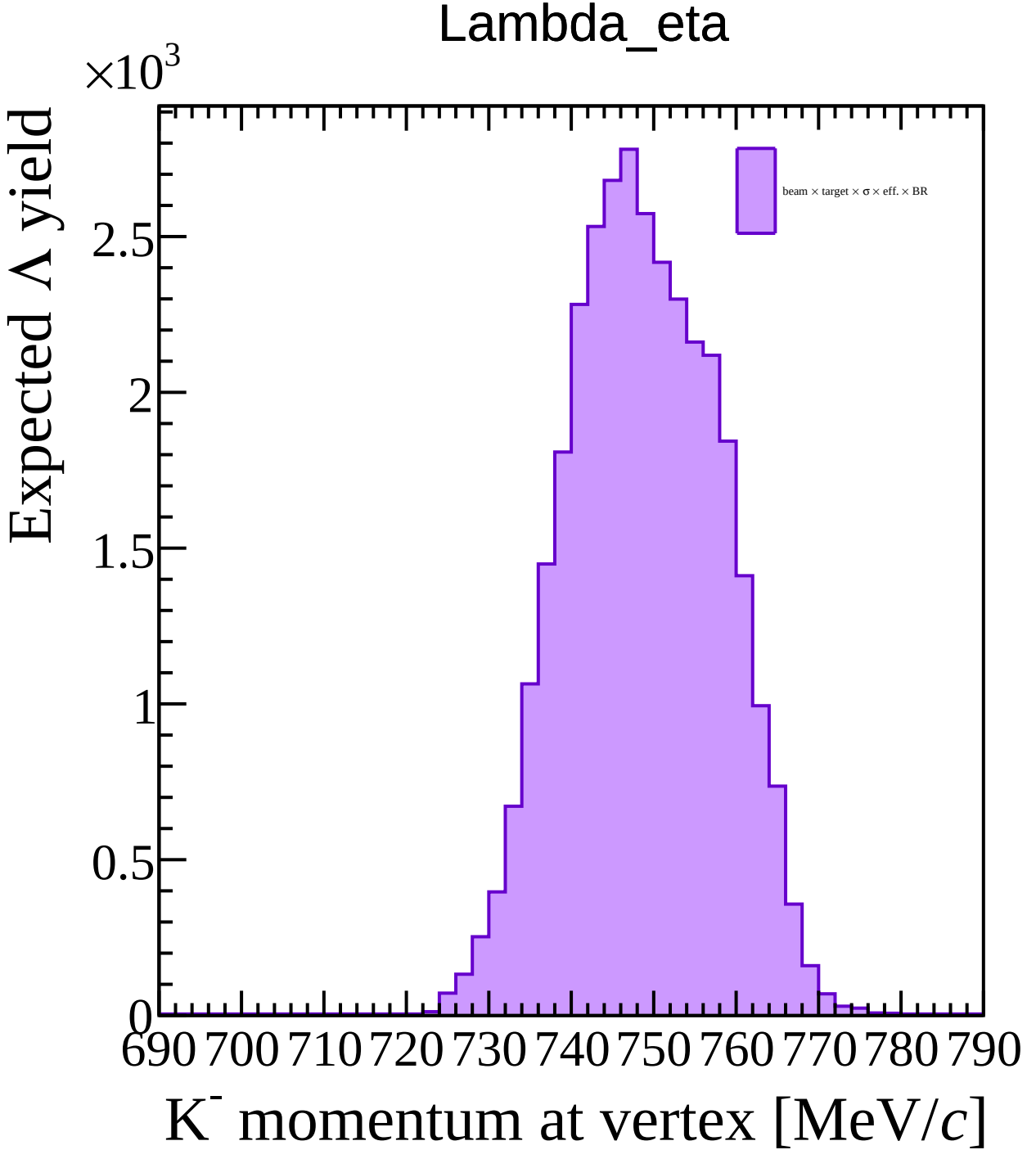
+ ntrack : 1.939e+04, + MM >= 0.5 : 2.444e+04, + ntrack + MM : 1.939e+04

LH2	1.262e+04
Kapton	1.216e+03
Mylar	4.983e+02
GFRP	8.726e+03
G10 top	8.955e+02
G10 bottom	4.866e+02


Lambda_eta
×10^3
Counts
reaction events
beam trigger total
physics trigger passed
K- momentum at vertex [MeV/c]


Lambda_eta
geometrical acceptance [%]
K- momentum at vertex [MeV/c]


Lambda_eta
total cross section [mb]
K- momentum at vertex [MeV/c]


Lambda_eta
×10^3
Expected Λ yield
beam × target × σ × eff. × BR
2.5
2
1.5
1
0.5
0
690 700 710 720 730 740 750 760 770 780 790
K⁻ momentum at vertex [MeV/c]

Lambda_pi0 : e72_tpcon-735_lambda_pi.root

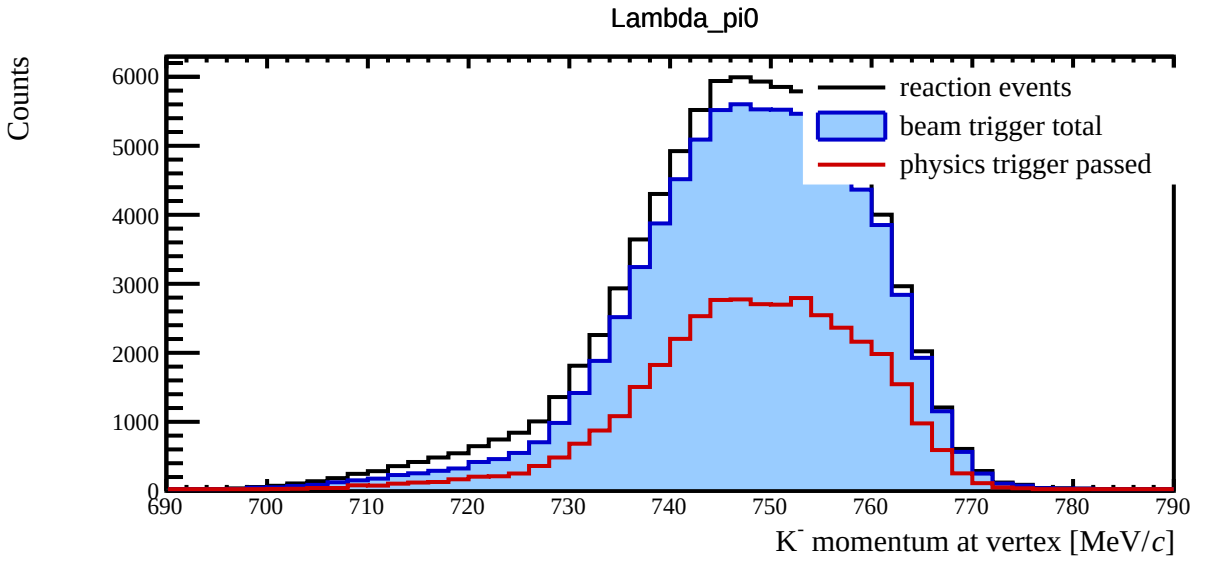
Trigger condition (reaction events)	count	% of beam
saved reaction events	89975	110.96
kaon beam	81087	100.00
beam x KVC veto	80391	99.14
x HTOF Mp >= 2	25819	31.84
x HTOF forward	35656	43.97
x HTOF Mp >= 2 and forward	21678	26.73
physics trigger (Mp >= 2 or forward)	39797	49.08
x p and pi- TPC hits	32542	40.13
x <= 1 other TPC track (ntrack)	32499	40.08
x MM >= 0.5 GeV/c2	0	0.00
x ntrack x MM >= 0.5 GeV/c2	0	0.00
x no other TPC track (ntrack0)	31517	38.87
x kinematics	5551	6.85
x ntrack x MM x kinematics	0	0.00

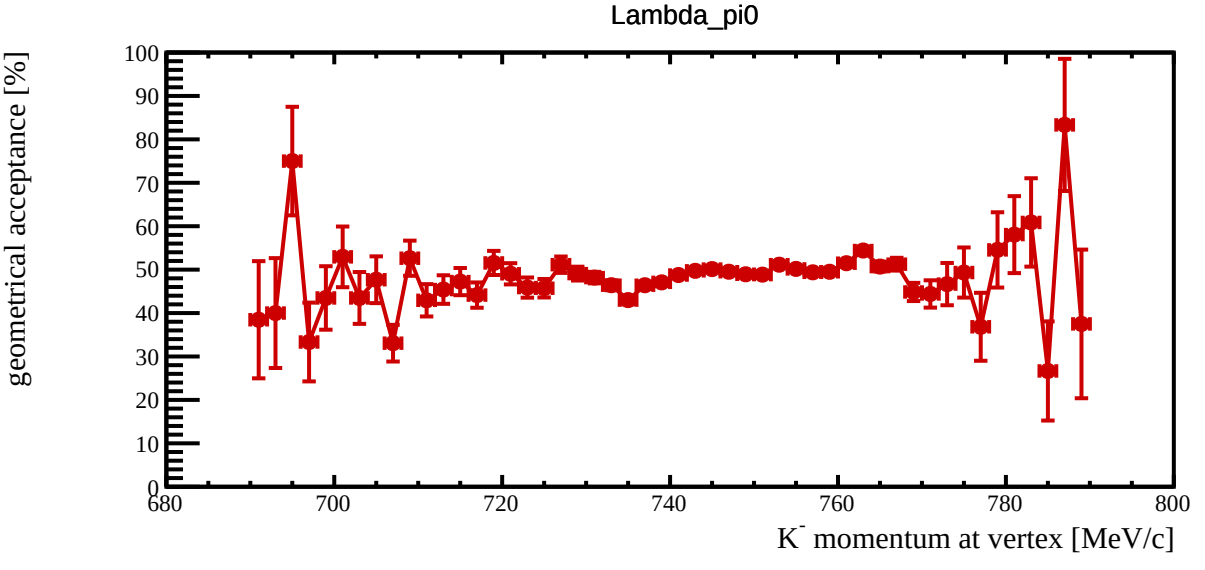
Generated reactions with beam trigger : 81088 (LH2 share 39246, 48.40 %)

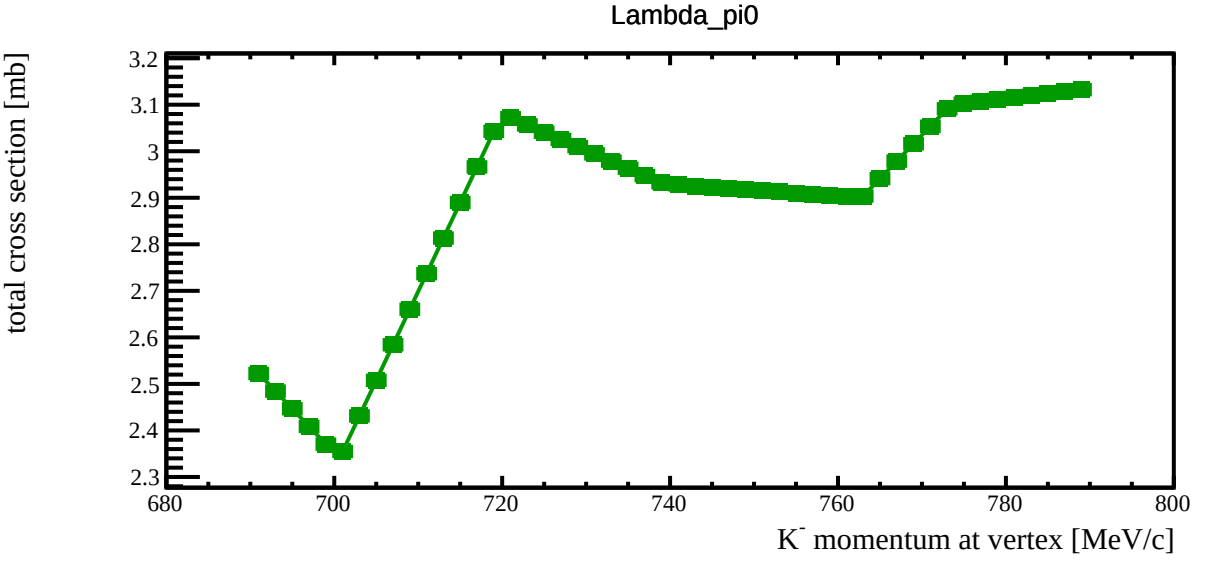
Vertex volume	beam trig.	phys. trig.	acc %	yield
LH2	39242	19415	49.48	3.010e+04
Kapton	3775	1915	50.73	2.970e+03
Mylar	1521	744	48.92	1.153e+03
GFRP	27569	13652	49.52	2.115e+04
G10 top	5651	2601	46.03	4.006e+03
G10 bottom	3329	1470	44.16	2.258e+03

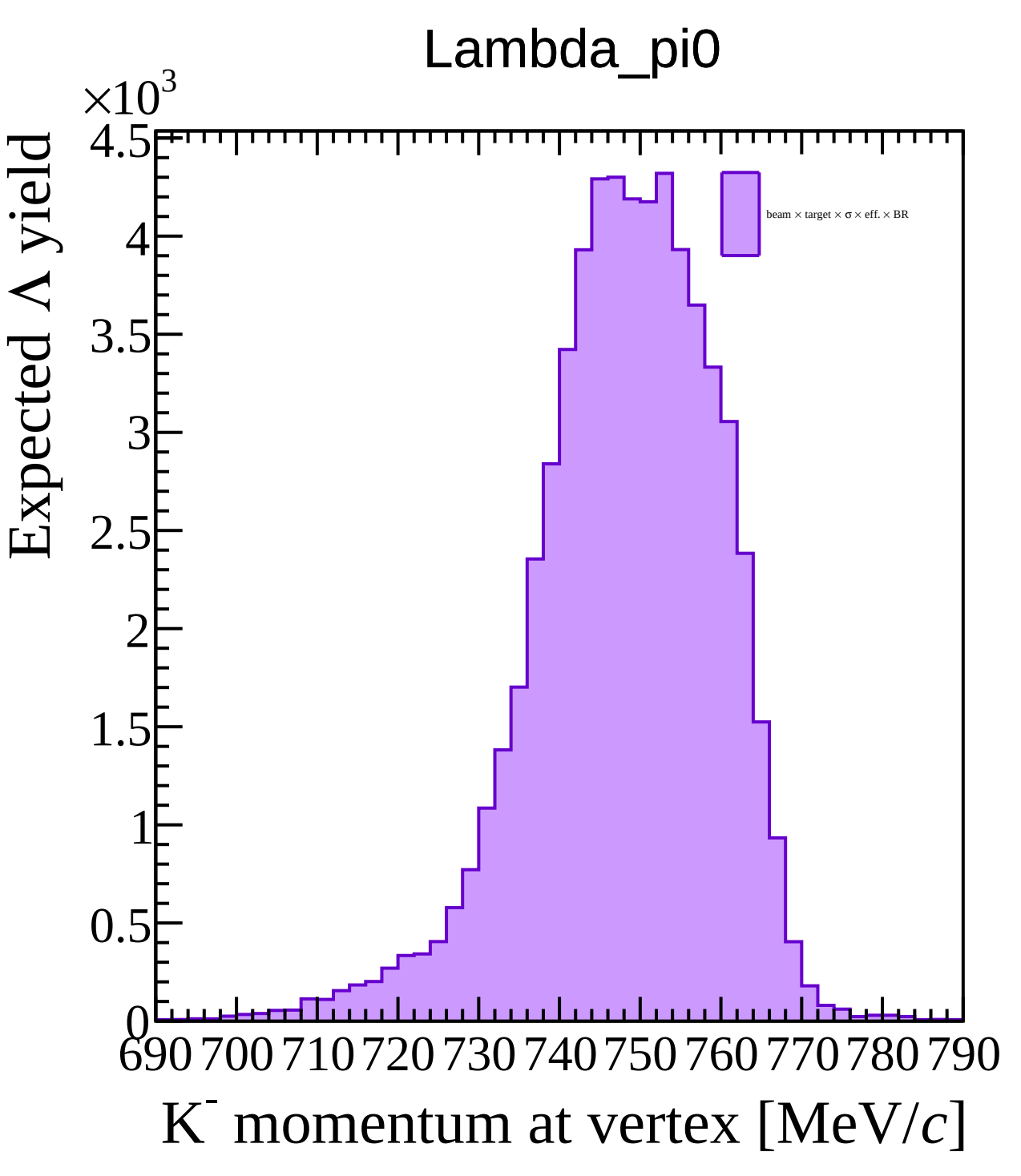
Expected Λ yield : 6.163e+04 (eff_tpc_L = 0.60)
  with p, pi- in >= 5 TPC layers : 5.042e+04 (also x eff_tpc_L)
  + ntrack : 5.036e+04, + MM >= 0.5 : 0.000e+00, + ntrack + MM : 0.000e+00

LH2	2.455e+04
Kapton	2.439e+03
Mylar	9.698e+02
GFRP	1.762e+04
G10 top	3.110e+03
G10 bottom	1.739e+03


Lambda_pi0
Counts
reaction events
beam trigger total
physics trigger passed
K- momentum at vertex [MeV/c]


Lambda_pi0
geometrical acceptance [%]
K- momentum at vertex [MeV/c]


Lambda_pi0
total cross section [mb]
K- momentum at vertex [MeV/c]


Lambda_pi0
×10^3
4.5
4
3.5
3
2.5
2
1.5
1
0.5
0
Expected Λ yield
beam × target × σ × eff. × BR
690 700 710 720 730 740 750 760 770 780 790
K⁻ momentum at vertex [MeV/c]

Sigma0_pi0 : e72_tpcon-735_sigma0_pi0.root

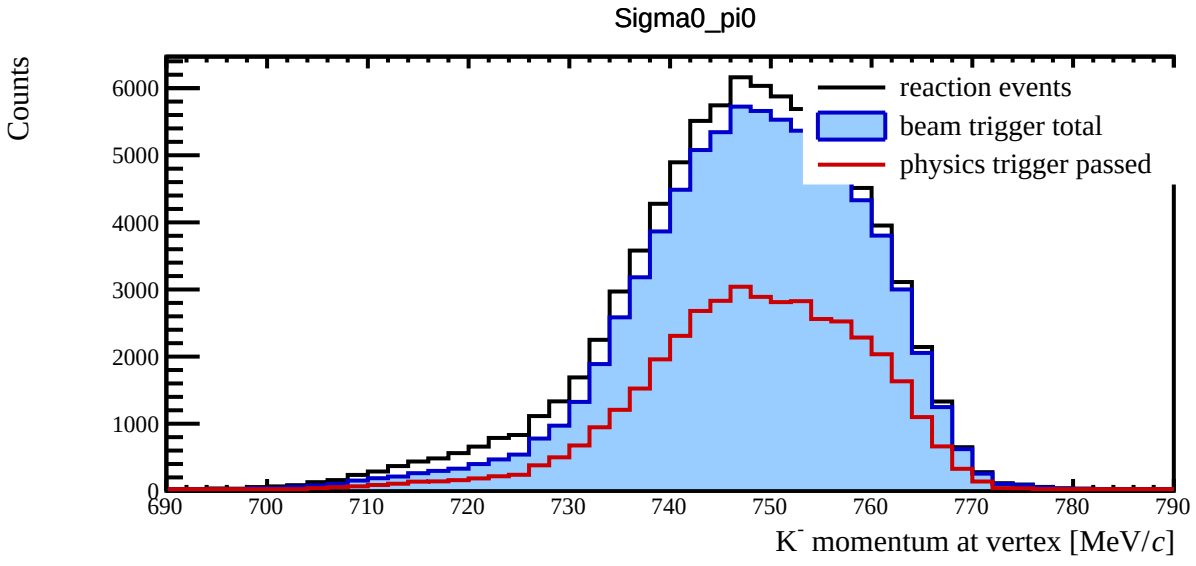
Trigger condition (reaction events)	count	% of beam
saved reaction events	90174	110.86
kaon beam	81340	100.00
beam x KVC veto	80799	99.33
x HTOF Mp >= 2	26530	32.62
x HTOF forward	38429	47.24
x HTOF Mp >= 2 and forward	23165	28.48
physics trigger (Mp >= 2 or forward)	41794	51.38
x p and pi- TPC hits	33775	41.52
x <= 1 other TPC track (ntrack)	33744	41.49
x MM >= 0.5 GeV/c2	0	0.00
x ntrack x MM >= 0.5 GeV/c2	0	0.00
x no other TPC track (ntrack0)	32703	40.21
x kinematics	7286	8.96
x ntrack x MM x kinematics	0	0.00

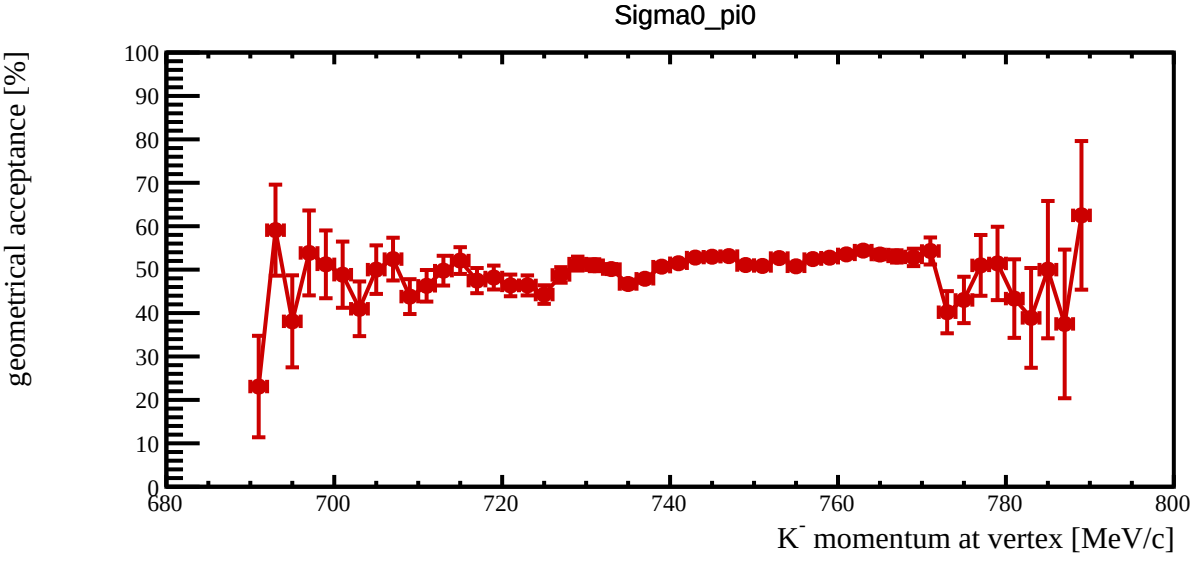
Generated reactions with beam trigger : 81340 (LH2 share 39307, 48.32 %)

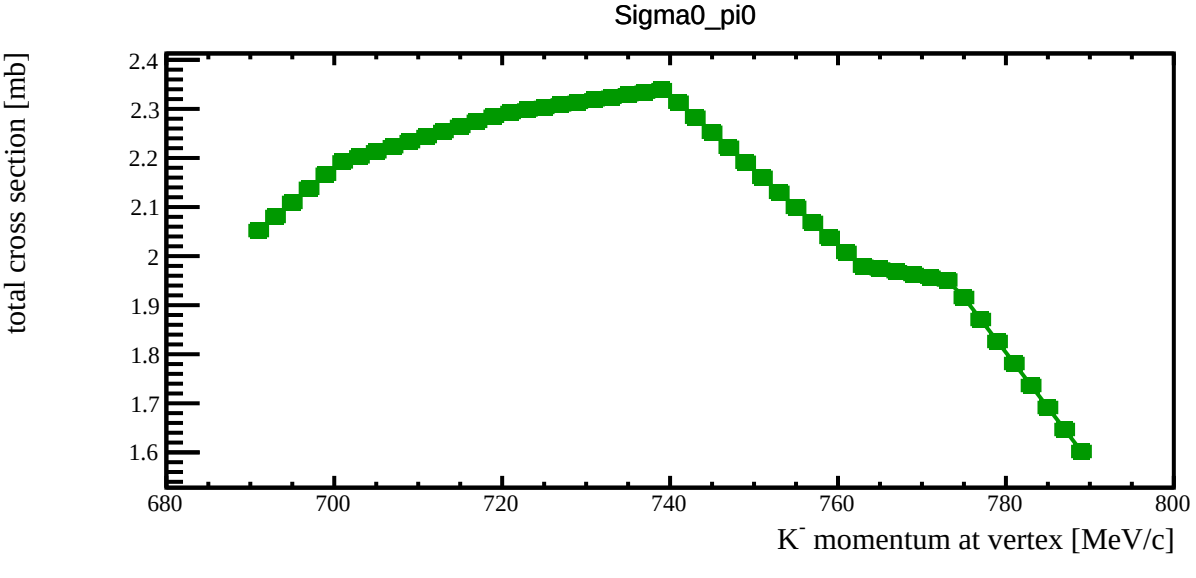
Vertex volume	beam trig.	phys. trig.	acc %	yield
LH2	39545	20496	51.83	2.355e+04
Kapton	3788	1948	51.43	2.236e+03
Al	1	1	100.00	1.043e+00
Mylar	1522	796	52.30	9.137e+02
GFRP	27470	14254	51.89	1.634e+04
G10 top	5785	2751	47.55	3.255e+03
G10 bottom	3229	1548	47.94	1.839e+03

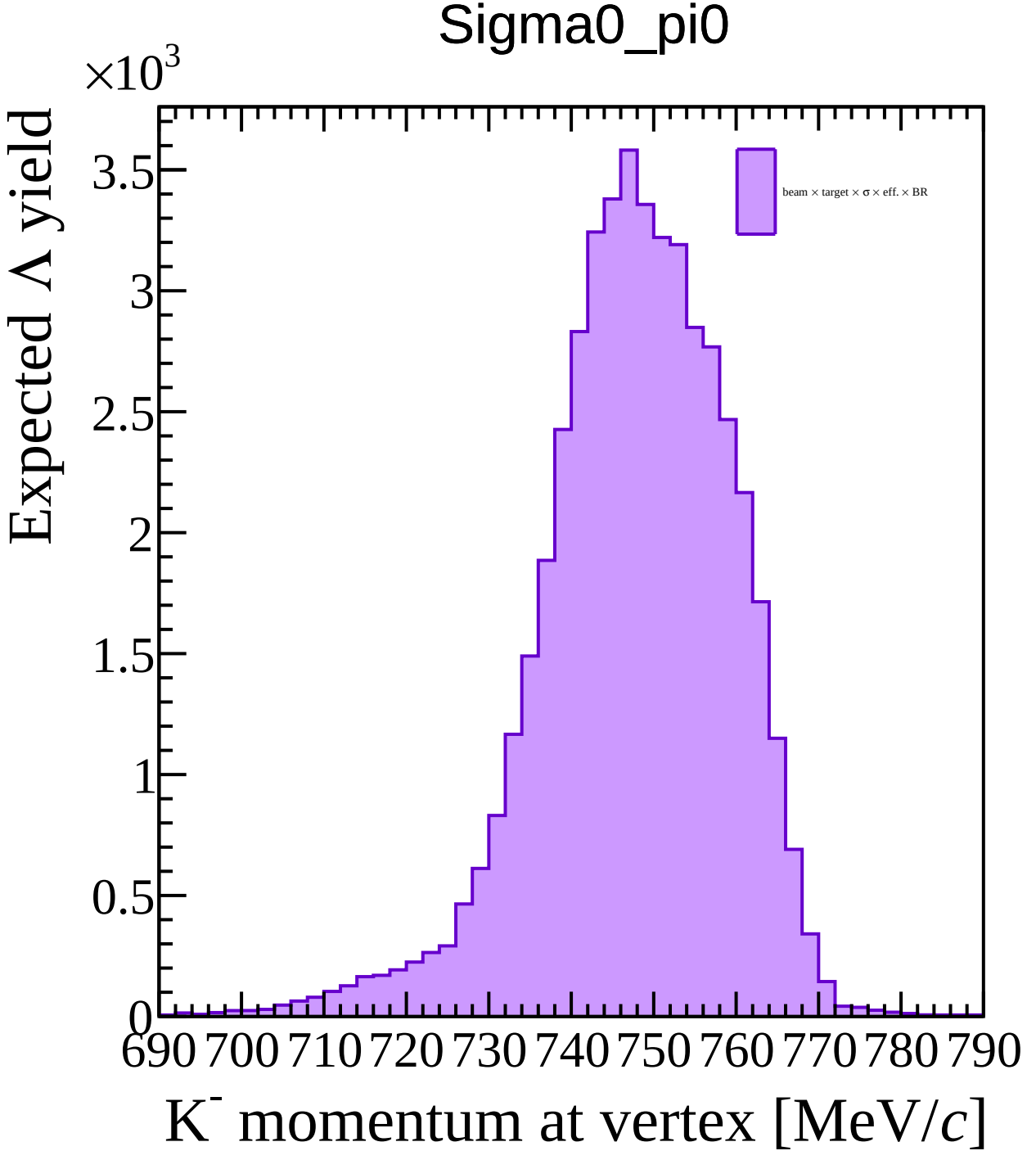
Expected Λ yield : 4.814e+04 (eff_tpc_L = 0.60)
  with p, pi- in >= 5 TPC layers : 3.890e+04 (also x eff_tpc_L)
  + ntrack : 3.886e+04, + MM >= 0.5 : 0.000e+00, + ntrack + MM : 0.000e+00

LH2	1.907e+04
Kapton	1.786e+03
Al	1.043e+00
Mylar	7.440e+02
GFRP	1.340e+04
G10 top	2.479e+03
G10 bottom	1.418e+03


Sigma0_pi0
Counts
reaction events
beam trigger total
physics trigger passed
6000
5000
4000
3000
2000
1000
0
690
700
710
720
730
740
750
760
770
780
790
K- momentum at vertex [MeV/c]


Sigma0_pi0
geometrical acceptance [%]
100
90
80
70
60
50
40
30
20
10
0
680
700
720
740
760
780
800
K- momentum at vertex [MeV/c]


Sigma0_pi0
total cross section [mb]
2.4
2.3
2.2
2.1
2
1.9
1.8
1.7
1.6
680
700
720
740
760
780
800
K- momentum at vertex [MeV/c]


Sigma0_pi0
×10^3
Expected Λ yield
3.5
3
2.5
2
1.5
1
0.5
0
beam × target × σ × eff. × BR
690 700 710 720 730 740 750 760 770 780 790
K⁻ momentum at vertex [MeV/c]

Sigmap_pim : e72_tpcon-735_sigmap_pim.root

Trigger condition (reaction events)	count	% of beam
saved reaction events	89775	111.14
kaon beam	80776	100.00
beam x KVC veto	79507	98.43
x HTOF Mp >= 2	43630	54.01
x HTOF forward	30932	38.29
x HTOF Mp >= 2 and forward	22181	27.46
physics trigger (Mp >= 2 or forward)	52381	64.85
x p and pi- TPC hits	28086	34.77
x <= 1 other TPC track (ntrack)	28039	34.71
x MM >= 0.5 GeV/c2	2	0.00
x ntrack x MM >= 0.5 GeV/c2	1	0.00
x no other TPC track (ntrack0)	27543	34.10
x kinematics	5	0.01
x ntrack x MM x kinematics	0	0.00

Generated reactions with beam trigger : 80776 (LH2 share 39061, 48.36 %)

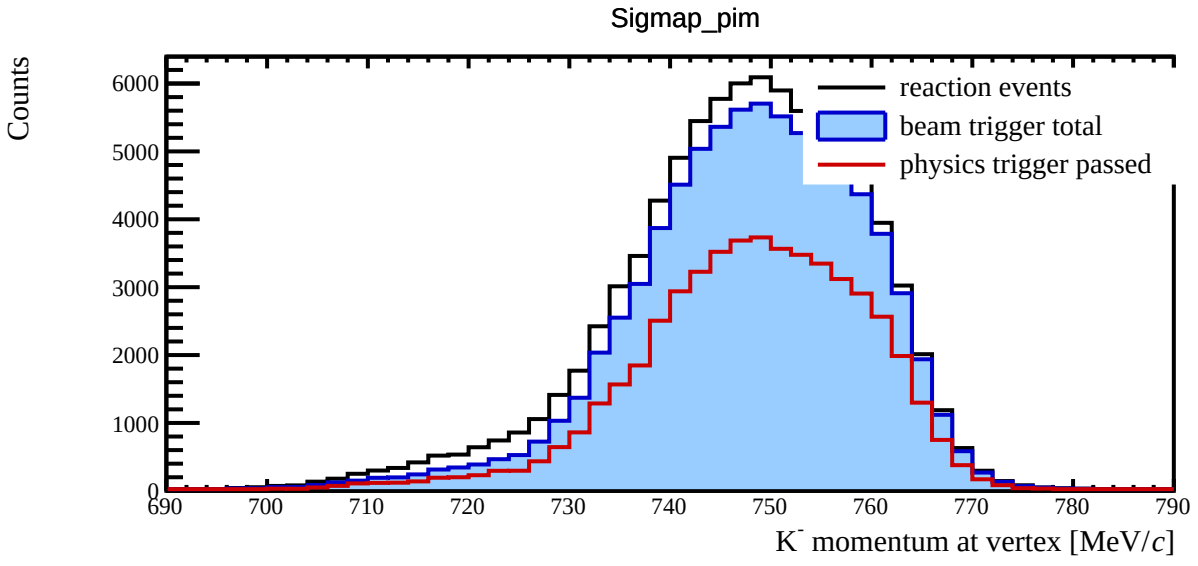
Vertex volume	beam trig.	phys. trig.	acc %	yield
LH2	38952	25451	65.34	2.557e+04
Kapton	3876	2572	66.36	2.580e+03
Mylar	1541	977	63.40	9.759e+02
GFRP	27343	18035	65.96	1.809e+04
G10 top	5845	3438	58.82	3.748e+03
G10 bottom	3219	1908	59.27	2.101e+03

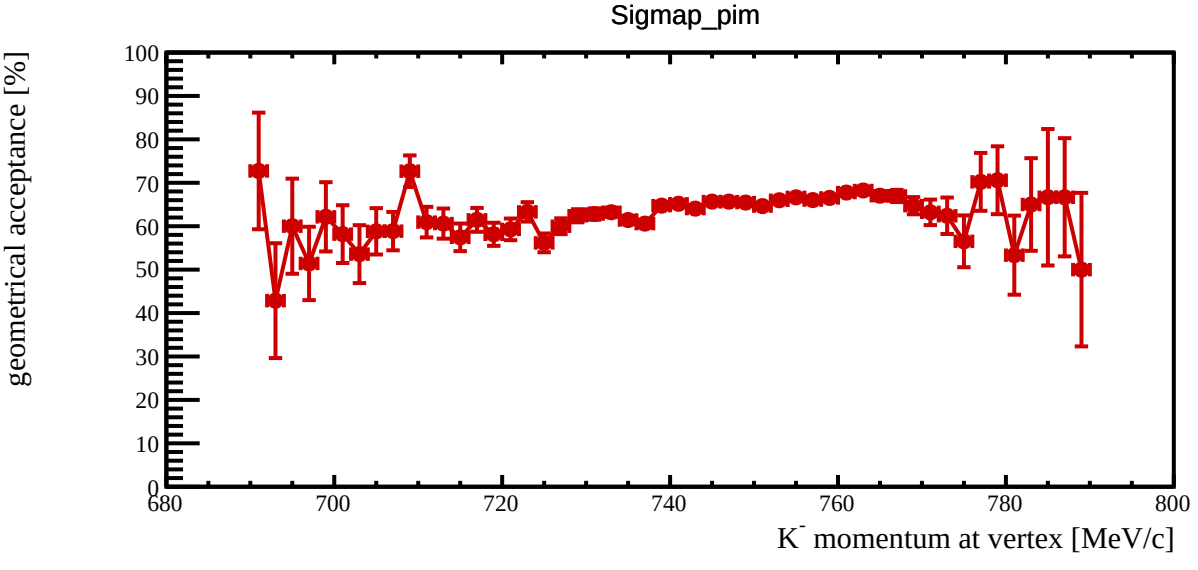
Expected Λ yield : 5.306e+04 (eff_tpc_L = 0.60)

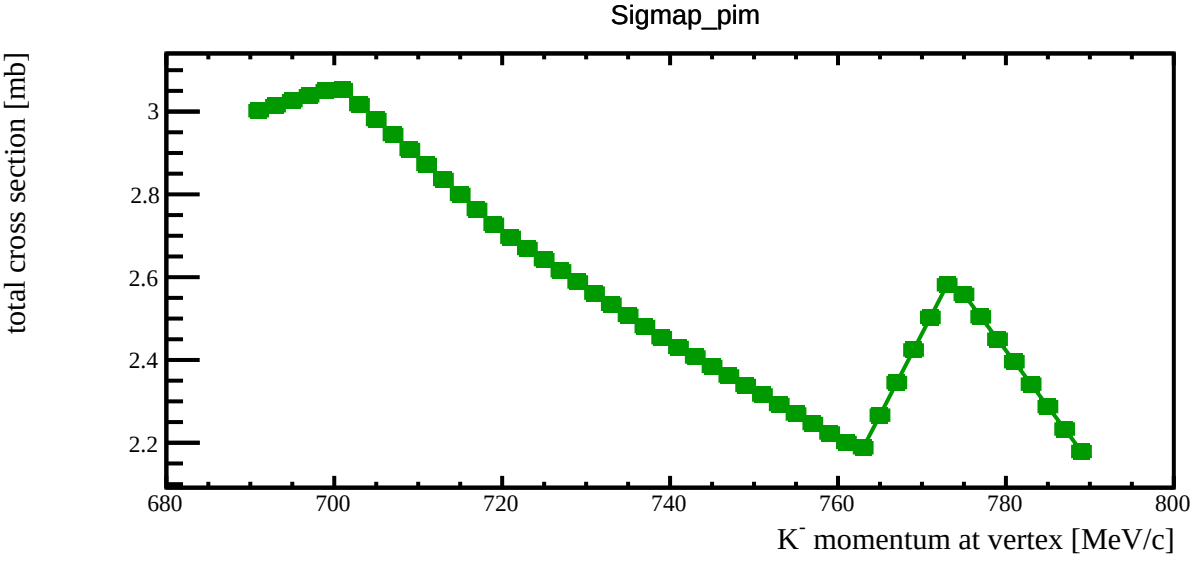
with p, pi- in >= 5 TPC layers : 2.843e+04 (also x eff_tpc_L)

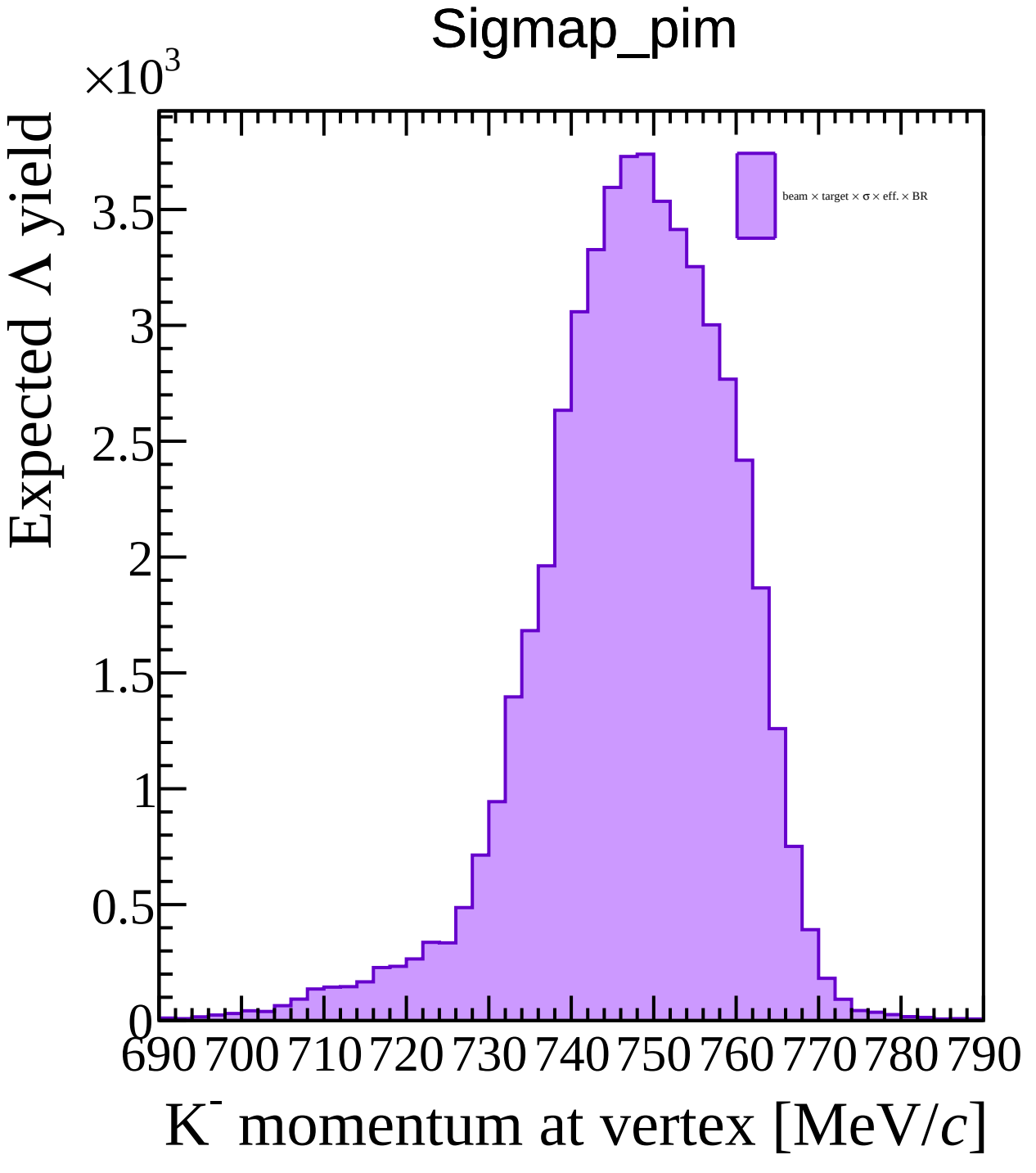
+ ntrack : 2.838e+04, + MM >= 0.5 : 2.199e+00, + ntrack + MM : 1.128e+00

LH2	1.387e+04
Kapton	1.425e+03
Mylar	5.196e+02
GFRP	9.785e+03
G10 top	1.832e+03
G10 bottom	1.002e+03


Sigmap_pim
Counts
reaction events
beam trigger total
physics trigger passed
6000
5000
4000
3000
2000
1000
0
690
700
710
720
730
740
750
760
770
780
790
K- momentum at vertex [MeV/c]


Sigmap_pim
geometrical acceptance [%]
100
90
80
70
60
50
40
30
20
10
0
680
700
720
740
760
780
800
K- momentum at vertex [MeV/c]


Sigmap_pim
total cross section [mb]
3
2.8
2.6
2.4
2.2
680
700
720
740
760
780
800
K- momentum at vertex [MeV/c]


Sigmap_pim
×10^3
Expected Λ yield
3.5
3
2.5
2
1.5
1
0.5
0
beam × target × σ × eff. × BR
690 700 710 720 730 740 750 760 770 780 790
K⁻ momentum at vertex [MeV/c]

Lambda_pip_pim : e72_tpcon-735_lambda_pi_pi.root

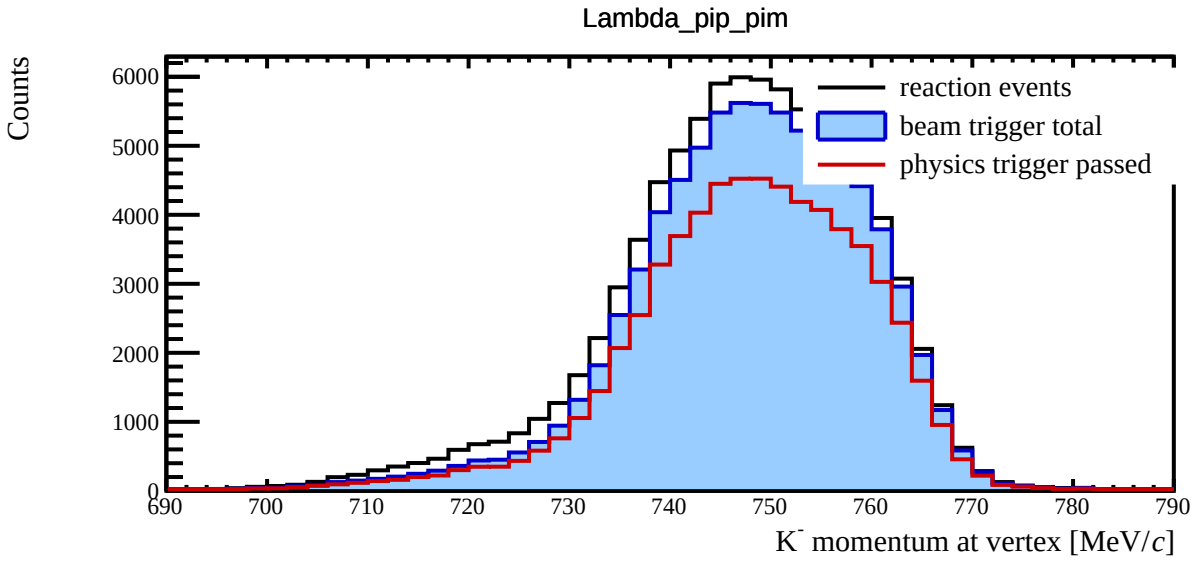
Trigger condition (reaction events)	count	% of beam
saved reaction events	89596	110.84
kaon beam	80833	100.00
beam x KVC veto	79220	98.00
x HTOF Mp >= 2	61845	76.51
x HTOF forward	49021	60.64
x HTOF Mp >= 2 and forward	45799	56.66
physics trigger (Mp >= 2 or forward)	65067	80.50
x p and pi- TPC hits	37948	46.95
x <= 1 other TPC track (ntrack)	6686	8.27
x MM >= 0.5 GeV/c2	5780	7.15
x ntrack x MM >= 0.5 GeV/c2	619	0.77
x no other TPC track (ntrack0)	271	0.34
x kinematics	13582	16.80
x ntrack x MM x kinematics	380	0.47

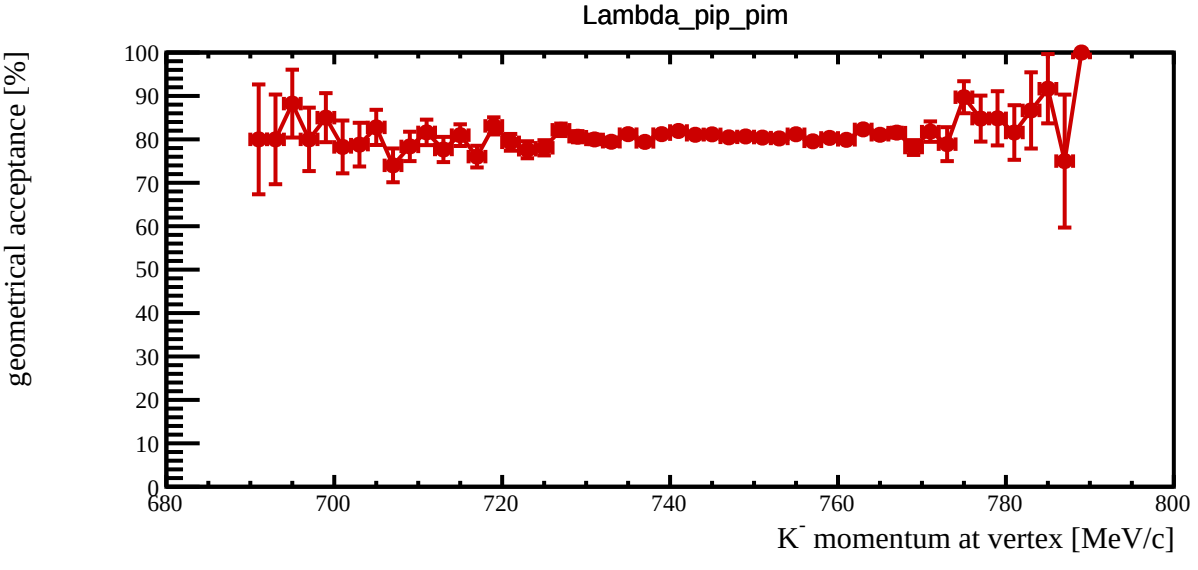
Generated reactions with beam trigger : 80833 (LH2 share 39107, 48.38 %)

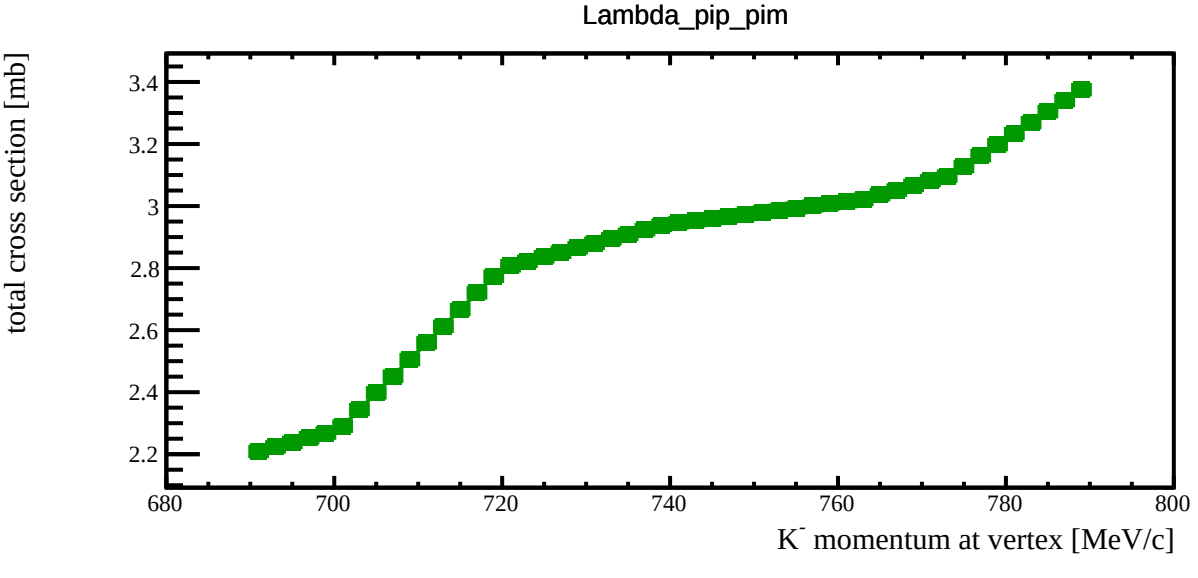
Vertex volume	beam trig.	phys. trig.	acc %	yield
LH2	39246	31760	80.93	5.019e+04
Kapton	3774	3066	81.24	4.847e+03
Al	1	1	100.00	1.571e+00
Mylar	1484	1209	81.47	1.910e+03
GFRP	27284	22007	80.66	3.473e+04
G10 top	5705	4435	77.74	6.628e+03
G10 bottom	3339	2589	77.54	3.828e+03

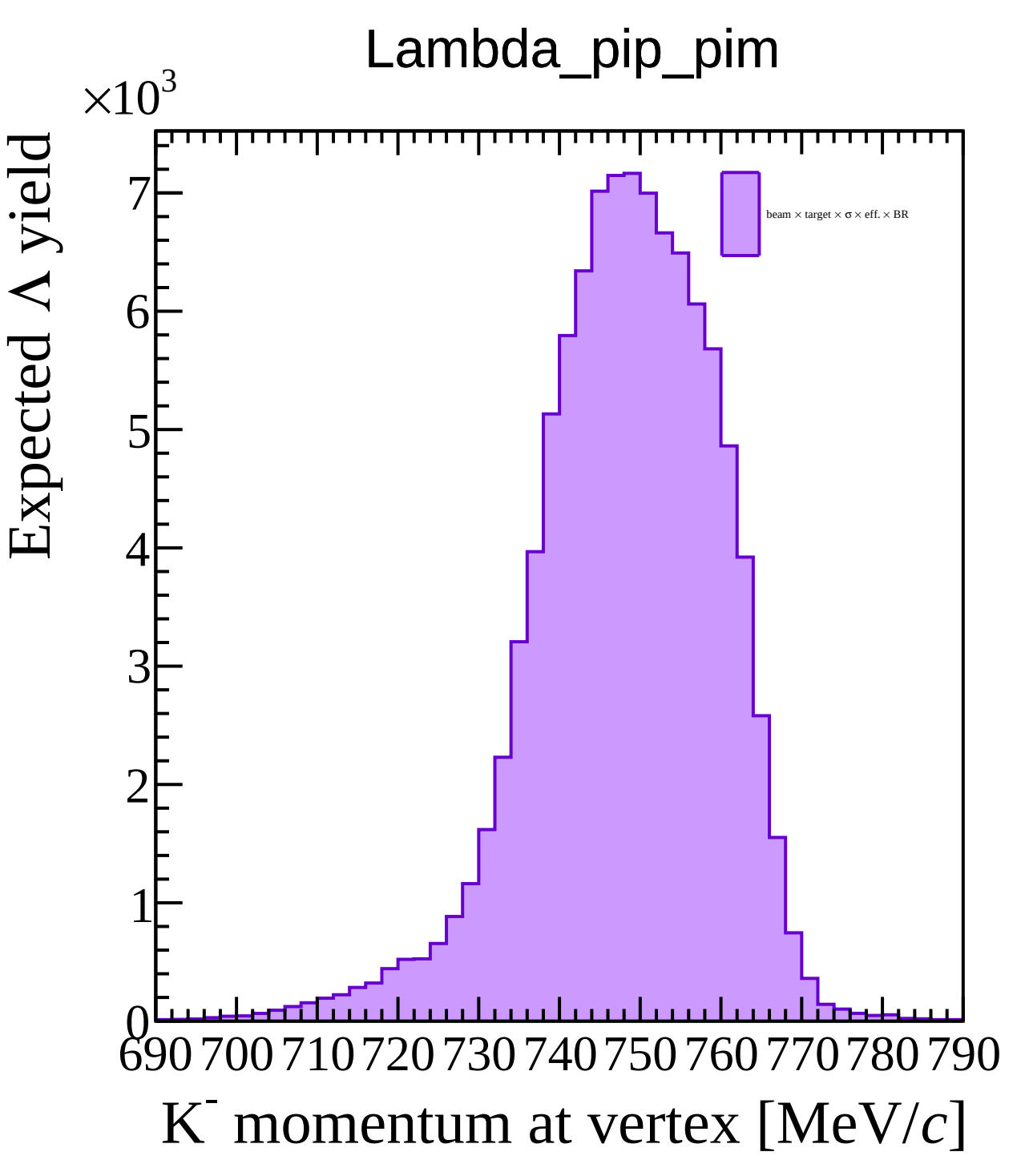
Expected Λ yield : 1.021e+05 (eff_tpc_L = 0.60)
  with p, pi- in >= 5 TPC layers : 5.967e+04 (also x eff_tpc_L)
  + ntrack : 1.047e+04, + MM >= 0.5 : 9.127e+03, + ntrack + MM : 9.777e+02

LH2	2.932e+04
Kapton	2.841e+03
Al	1.571e+00
Mylar	1.116e+03
GFRP	2.061e+04
G10 top	3.618e+03
G10 bottom	2.165e+03


Lambda_pip_pim
Counts
reaction events
beam trigger total
physics trigger passed
K- momentum at vertex [MeV/c]


Lambda_pip_pim
geometrical acceptance [%]
K- momentum at vertex [MeV/c]


Lambda_pip_pim
total cross section [mb]
K- momentum at vertex [MeV/c]


Lambda_pip_pim
×10^3
Expected Λ yield
beam × target × σ × eff. × BR
0
1
2
3
4
5
6
7
690
700
710
720
730
740
750
760
770
780
790
K⁻ momentum at vertex [MeV/c]

Sigma0_pip_pim : e72_tpcon-735_sigma0_pip_pim.root

Trigger condition (reaction events)	count	% of beam
saved reaction events	89966	111.01
kaon beam	81045	100.00
beam x KVC veto	79414	97.99
x HTOF Mp >= 2	61383	75.74
x HTOF forward	52195	64.40
x HTOF Mp >= 2 and forward	48289	59.58
physics trigger (Mp >= 2 or forward)	65289	80.56
x p and pi- TPC hits	38545	47.56
x <= 1 other TPC track (ntrack)	7959	9.82
x MM >= 0.5 GeV/c2	12932	15.96
x ntrack x MM >= 0.5 GeV/c2	2178	2.69
x no other TPC track (ntrack0)	376	0.46
x kinematics	16988	20.96
x ntrack x MM x kinematics	1393	1.72

Generated reactions with beam trigger : 81252 (LH2 share 39284, 48.35 %)

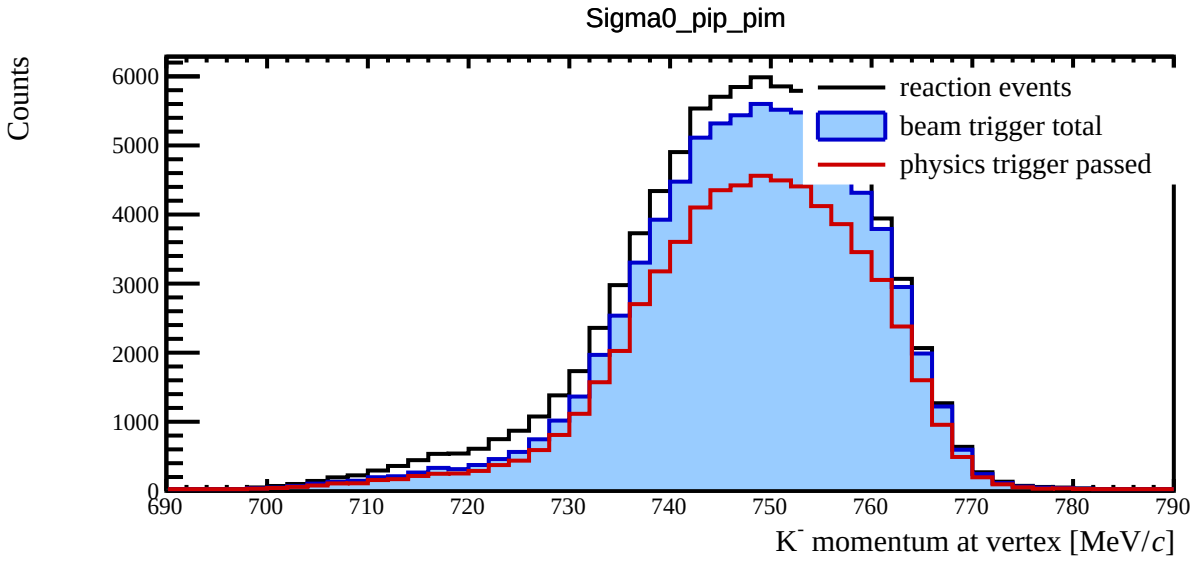
Vertex volume	beam trig.	phys. trig.	acc %	yield
LH2	39234	31764	80.96	1.027e+04
Kapton	3777	3023	80.04	9.780e+02
Mylar	1521	1252	82.31	4.056e+02
GFRP	27540	22334	81.10	7.206e+03
G10 top	5665	4320	76.26	1.260e+03
G10 bottom	3308	2596	78.48	7.402e+02

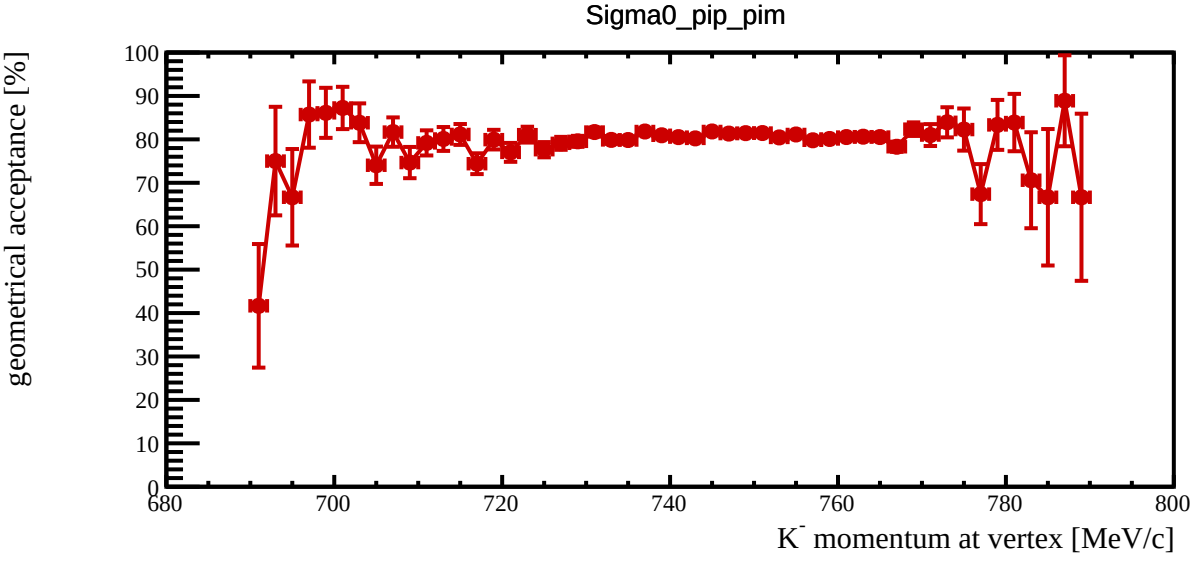
Expected Λ yield : 2.086e+04 (eff_tpc_L = 0.60)

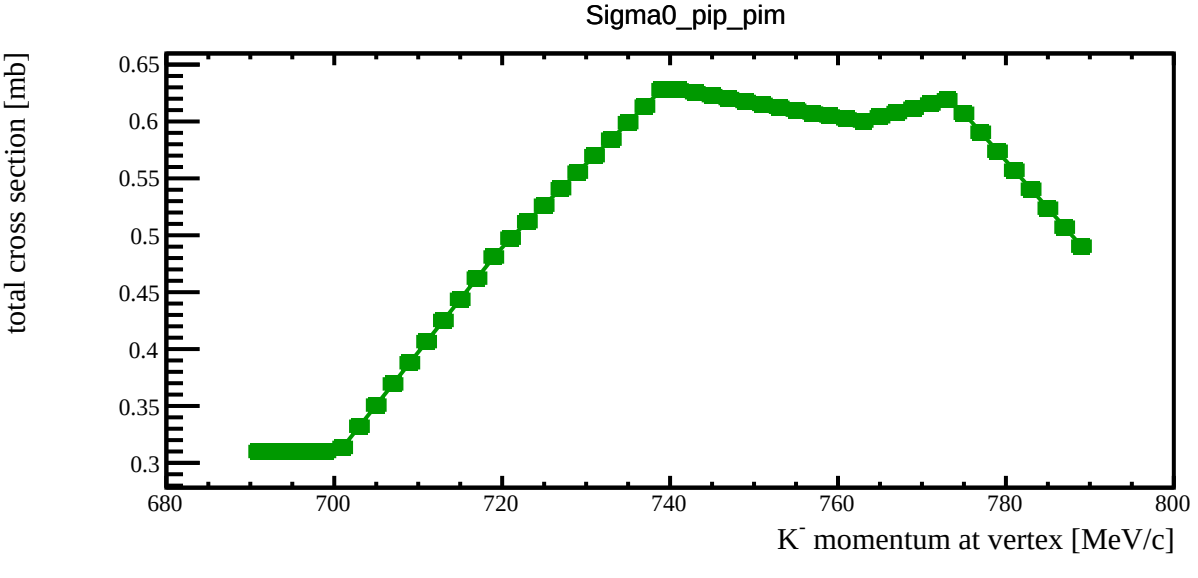
with p, pi- in >= 5 TPC layers : 1.233e+04 (also x eff_tpc_L)

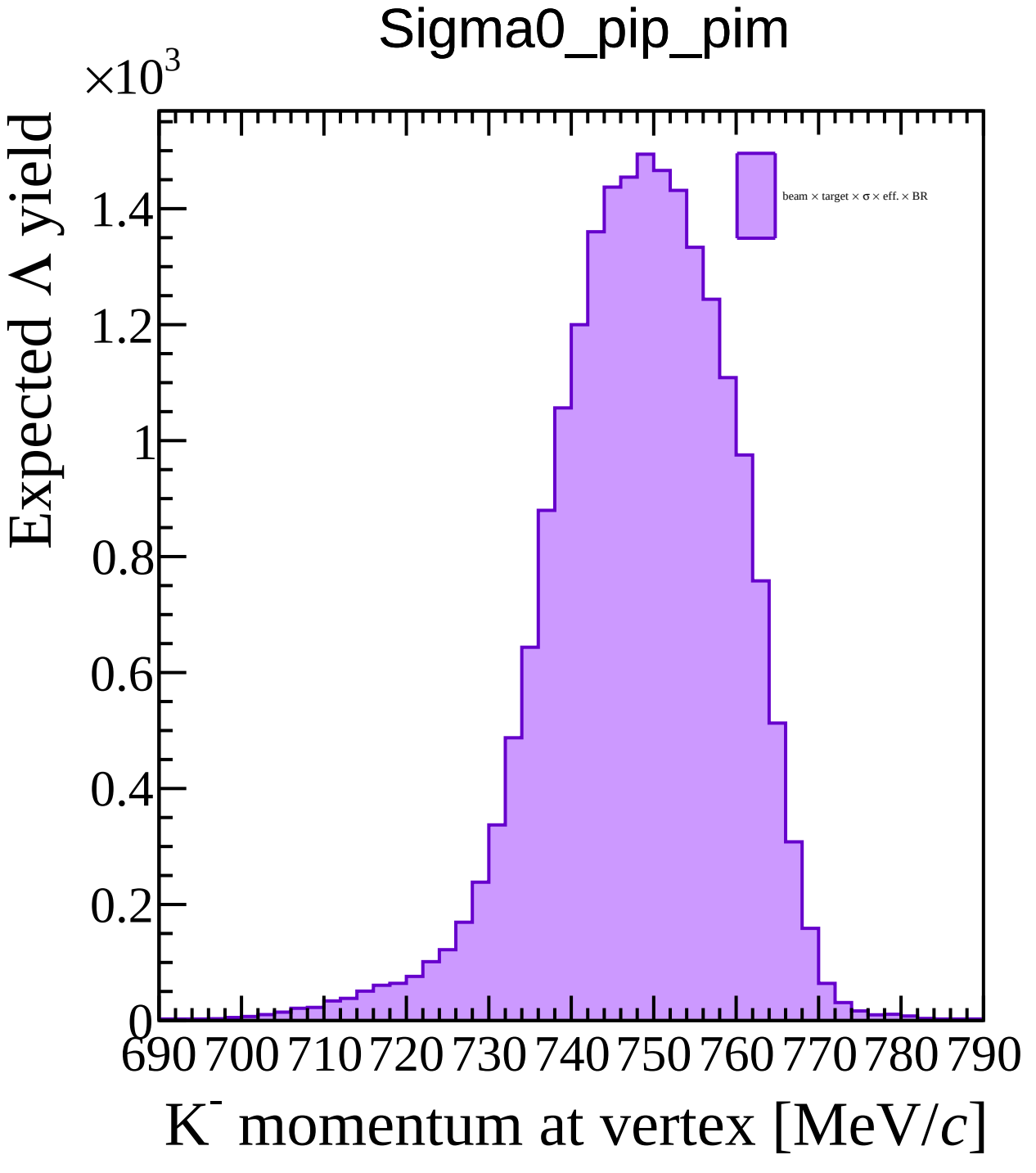
+ ntrack : 2.530e+03, + MM >= 0.5 : 4.159e+03, + ntrack + MM : 6.991e+02

LH2	6.091e+03
Kapton	5.848e+02
Mylar	2.496e+02
GFRP	4.323e+03
G10 top	6.830e+02
G10 bottom	3.992e+02


Sigma0_pip_pim
Counts
reaction events
beam trigger total
physics trigger passed
K- momentum at vertex [MeV/c]


Sigma0_pip_pim
geometrical acceptance [%]
K- momentum at vertex [MeV/c]


Sigma0_pip_pim
total cross section [mb]
K- momentum at vertex [MeV/c]


Sigma0_pip_pim
×10^3
Expected Λ yield
1.4
1.2
1
0.8
0.6
0.4
0.2
0
beam × target × σ × eff. × BR
690 700 710 720 730 740 750 760 770 780 790
K⁻ momentum at vertex [MeV/c]

Sigmap_pim_pi0 : e72_tpcon-735_sigmap_pim_pi0.root

Trigger condition (reaction events)	count	% of beam
saved reaction events	89496	110.85
kaon beam	80733	100.00
beam x KVC veto	79783	98.82
x HTOF Mp >= 2	42736	52.93
x HTOF forward	39778	49.27
x HTOF Mp >= 2 and forward	26631	32.99
physics trigger (Mp >= 2 or forward)	55883	69.22
x p and pi- TPC hits	29019	35.94
x <= 1 other TPC track (ntrack)	28985	35.90
x MM >= 0.5 GeV/c2	1	0.00
x ntrack x MM >= 0.5 GeV/c2	1	0.00
x no other TPC track (ntrack0)	28211	34.94
x kinematics	12	0.01
x ntrack x MM x kinematics	1	0.00

Generated reactions with beam trigger : 80903 (LH2 share 39044, 48.26 %)

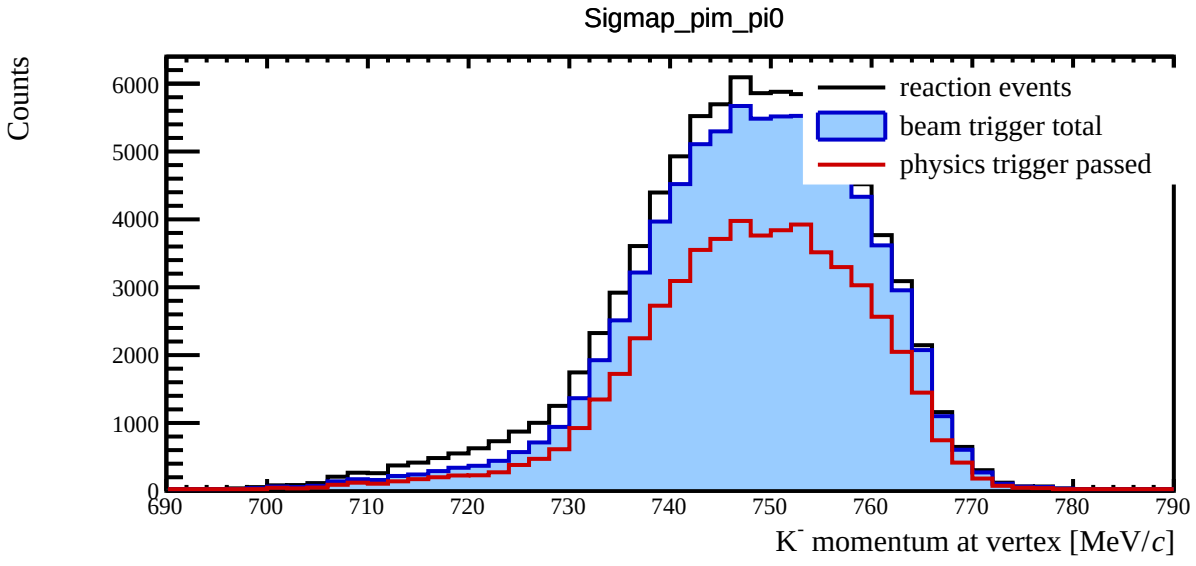
Vertex volume	beam trig.	phys. trig.	acc %	yield
LH2	38881	27234	70.04	8.553e+03
Kapton	3805	2707	71.14	8.499e+02
Mylar	1535	1099	71.60	3.439e+02
GFRP	27572	19293	69.97	6.047e+03
G10 top	5696	3545	62.24	9.299e+02
G10 bottom	3244	2005	61.81	5.105e+02

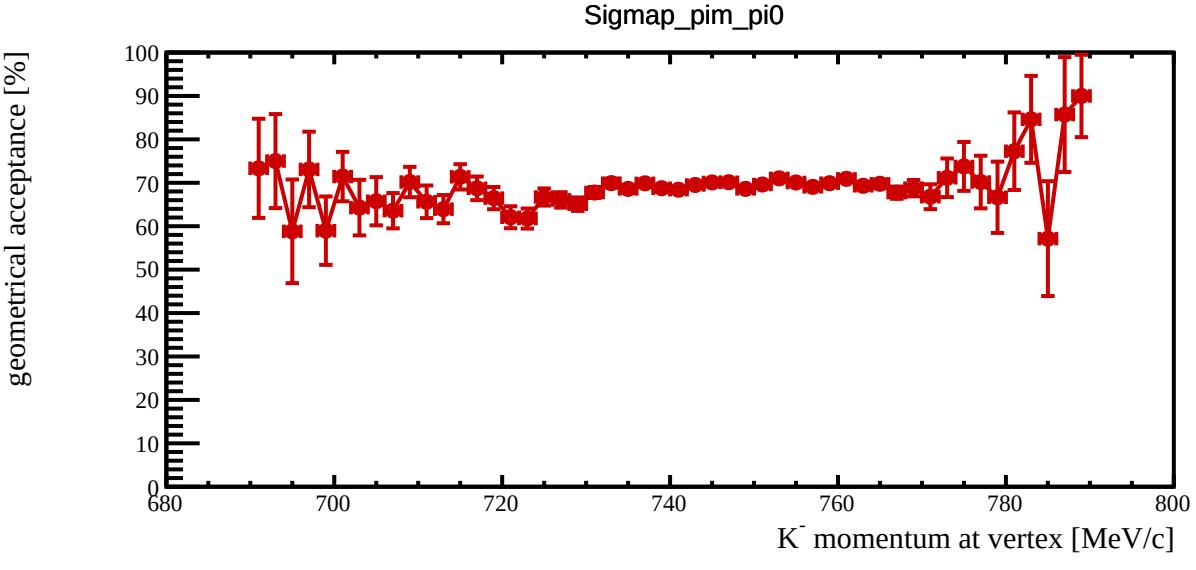
Expected Λ yield : 1.723e+04 (eff_tpc_L = 0.60)

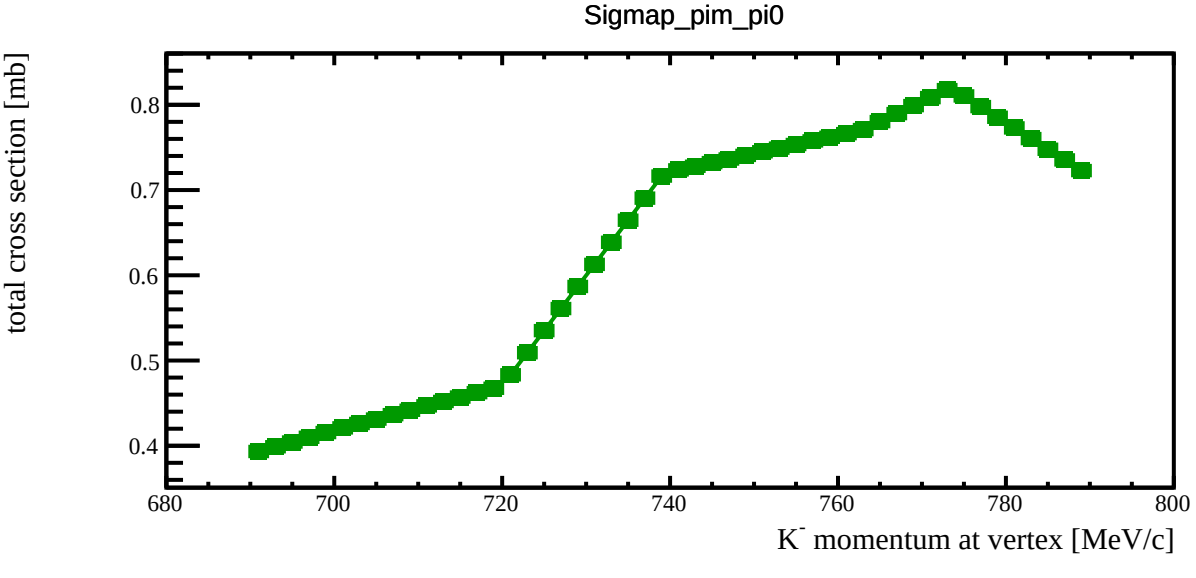
with p, pi- in >= 5 TPC layers : 8.973e+03 (also x eff_tpc_L)

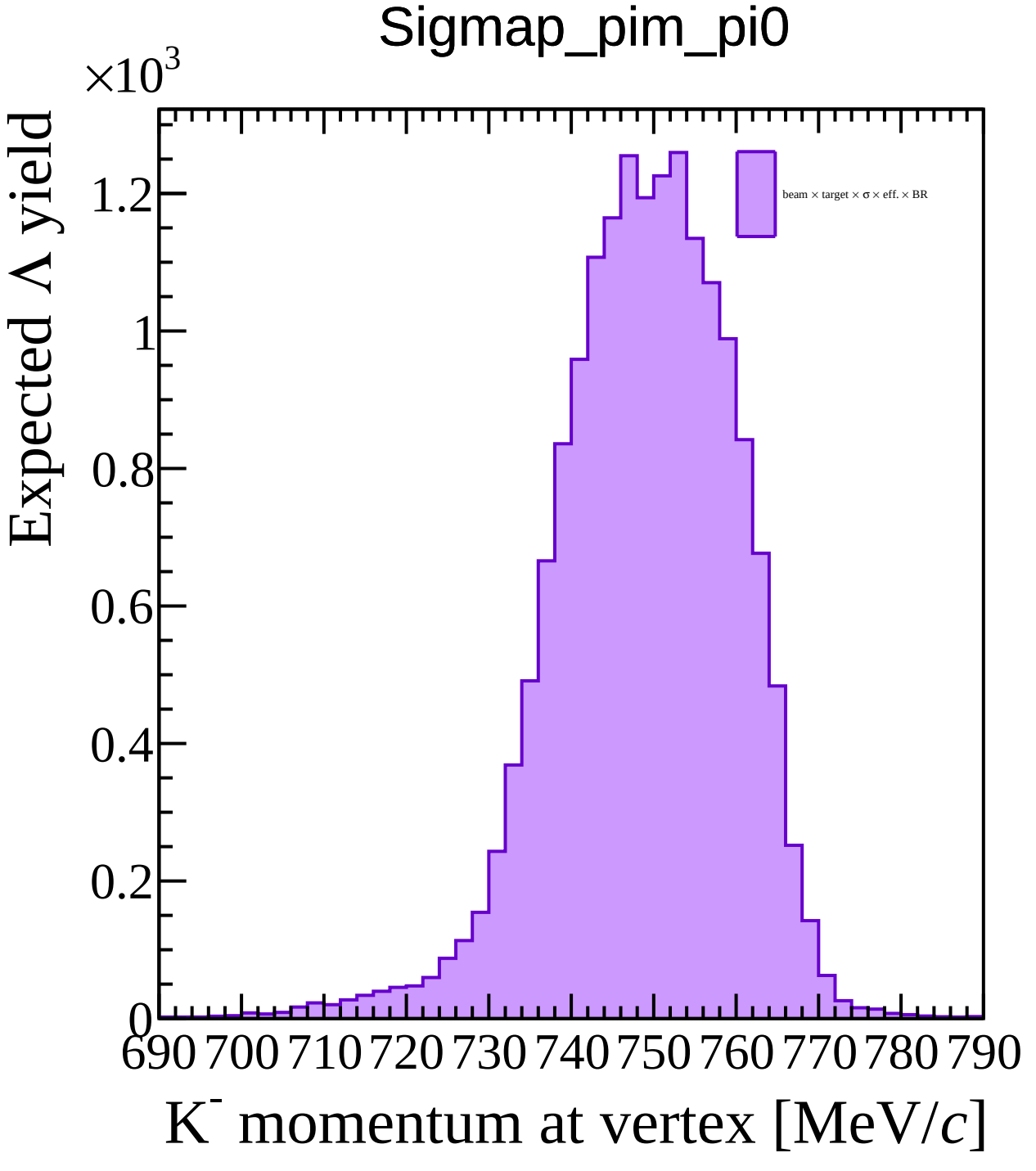
+ ntrack : 8.962e+03, + MM >= 0.5 : 3.215e-01, + ntrack + MM : 3.215e-01

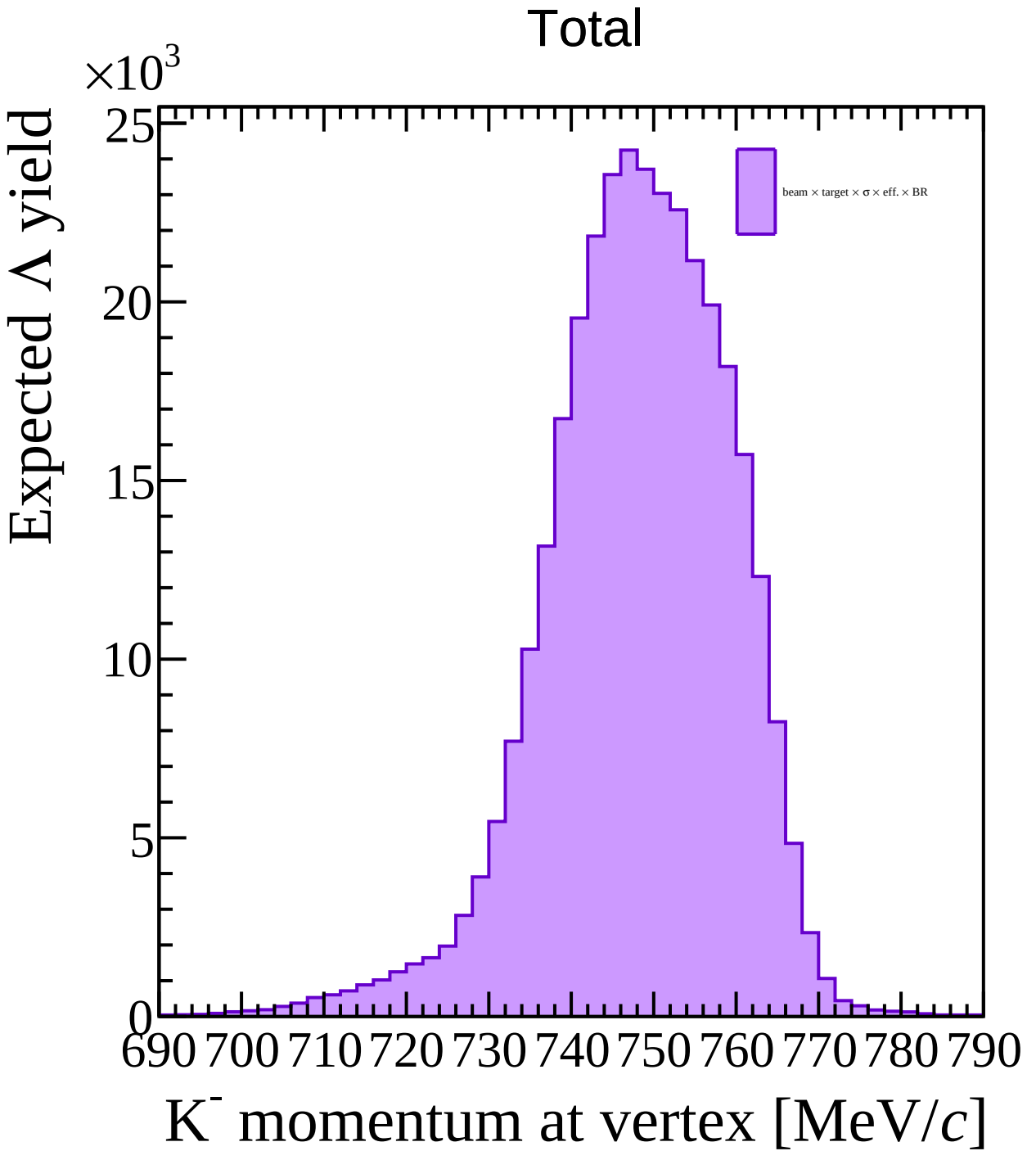
LH2	4.484e+03
Kapton	4.652e+02
Mylar	1.835e+02
GFRP	3.181e+03
G10 top	4.242e+02
G10 bottom	2.351e+02


Sigmap_pim_pi0
Counts
reaction events
beam trigger total
physics trigger passed
6000
5000
4000
3000
2000
1000
0
690
700
710
720
730
740
750
760
770
780
790
K- momentum at vertex [MeV/c]


Sigmap_pim_pi0
geometrical acceptance [%]
100
90
80
70
60
50
40
30
20
10
0
680
700
720
740
760
780
800
K- momentum at vertex [MeV/c]


Sigmap_pim_pi0
total cross section [mb]
0.8
0.7
0.6
0.5
0.4
680
700
720
740
760
780
800
K- momentum at vertex [MeV/c]


Sigmap_pim_pi0
×10^3
Expected Λ yield
1.2
1
0.8
0.6
0.4
0.2
0
beam × target × σ × eff. × BR
690 700 710 720 730 740 750 760 770 780 790
K⁻ momentum at vertex [MeV/c]


Total
×10^3
25
20
15
10
5
0
Expected Λ yield
beam × target × σ × eff. × BR
690 700 710 720 730 740 750 760 770 780 790
K⁻ momentum at vertex [MeV/c]